

Stage 2 Validation Architecture - Media Delivery Pipeline

V2

Purpose: validate artifact labels, format transitions, protocol labels, handoff teaching, and why-here teaching without enterprise multi-rail complexity.

Important Test-Data Change

This V2 avoids ambiguous labels such as Content Provider and repeated Service suffixes. The first node is now Provider Source, and downstream nodes use distinct labels so extraction should not collapse the journey into generic Service.

Architecture Flow

Provider Source

| MP2

v

Ingest Gateway

| MP2

v

Transcoder

| H.264

v

Packager

| MPD

v

Origin Store

| HTTPS

v

Edge Cache

| HTTPS

v

Player App

Numbered Steps

1. Provider Source uploads MP2 source media to Ingest Gateway.
2. Ingest Gateway forwards MP2 source media to Transcoder.
3. Transcoder converts source media into H.264 delivery renditions.
4. Packager prepares MPD manifests for downstream delivery.
5. Origin Store holds packaged delivery artifacts for downstream retrieval.
6. Edge Cache retrieves packaged delivery artifacts from Origin Store over HTTPS.
7. Player App requests content through Edge Cache over HTTPS.

Glossary

MP2 = MPEG Transport Stream
H.264 = Video codec format
MPD = DASH Manifest
HTTPS = Web transport protocol
Provider Source = External media source for the documented journey
Ingest Gateway = Entry point for source media in this document
Transcoder = Component that converts media formats
Packager = Component that prepares delivery artifacts
Origin Store = Component that holds packaged delivery artifacts
Edge Cache = Delivery cache near viewers
Player App = Viewer-side playback client

Operational Notes

Transcoder reduces playback compatibility issues.
Packager prepares delivery artifacts for downstream consumption.
Origin Store provides a centralized delivery source in this document.
Edge Cache reduces latency and origin load.

Expected Validation Signals

Area	Expected Signal
Artifact Detection	MP2, H.264, MPD, HTTPS should appear as artifacts or public labels.
Handoff Artifact Mapping	MP2 should appear near early hops, MPD near Packager to Origin Store, HTTPS near Origin
Narration	Rail narration may say: The document identifies MPD as a manifest artifact associated with t
Extractor Check	Canonical path should start with Provider Source, not generic Service.